THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL UNIT TEST SET 3

Chapters 12–14: **Surface Areas and Volumes, Statistics, Probability** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A solid cylinder of radius 7 cm and height 8 cm is melted and recast into a cone of the same base radius. The height of the cone is: 1

(a) 8 cm **(b)** 16 cm **(c)** 24 cm **(d)** 48 cm

2. For the data with classes 200–220, 220–240, 240–260, 260–280, 280–300, 300–320, 320–340 with frequencies 10, 22, 24, 6, 21, 2, 15 respectively, The mode of the data is: 1

(a) 242 **(b)** 264.4 **(c)** 240 **(d)** 260

3. A bag contains 11 red and 9 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{16}{25}$. The value of x is: 1

(a) 6 **(b)** 10 **(c)** 4 **(d)** 5

4. A solid cone and a solid hemisphere have equal base radii 10 cm and equal volumes. The height of the cone is: 1

(a) 10 cm **(b)** 15 cm **(c)** 20 cm **(d)** 30 cm

5. The modal class is 20–30 with $f_1 = 16$, $f_0 = 6$, $f_2 = 8$. The mode is: 1

(a) $20 + \frac{10}{16} \times 10$ **(b)** $20 + \frac{10}{22} \times 10$ **(c)** $20 + \frac{10}{18} \times 10$ **(d)** $20 + \frac{8}{18} \times 10$

6. A bag contains 13 red and 7 blue balls. 7 more red balls are put into the bag and a ball is drawn at random. The probability that the ball drawn is red is: 1

(a) $\frac{19}{27}$ **(b)** $\frac{7}{27}$ **(c)** $\frac{7}{9}$ **(d)** $\frac{20}{27}$

7. The radii of two solid spheres are in the ratio 3 : 5. The ratio of their surface areas is: 1

(a) 18 : 25 **(b)** 9 : 25 **(c)** 27 : 125 **(d)** 3 : 5

8. The mean of a distribution with classes 40–60, 60–80, 80–100, 100–120, 120–140, 140–160, 160–180, 180–200 and frequencies 22, 12, 13, p, 5, 2, 8, 13 respectively is 106.4. The value of p is: 1

(a) 30 **(b)** 25 **(c)** 24 **(d)** 27

9. A box contains 80 tickets numbered 1 to 80. One ticket is drawn at random. The probability that the number drawn is a multiple of 3 or 5 is: 1

(a) $\frac{19}{40}$ **(b)** $\frac{37}{80}$ **(c)** $\frac{9}{20}$ **(d)** $\frac{43}{80}$

10. The radii of two solid spheres are in the ratio 3 : 4. The ratio of their volumes is: 1

(a) 27 : 64 **(b)** 3 : 4 **(c)** 9 : 16 **(d)** 27 : 32

11. The median of a distribution with classes 20–40, 40–60, 60–80, 80–100, 100–120, 120–140, 140–160 and frequencies 20, x, 16, y, 21, 5, 7 respectively, and total frequency 100, is 70. The value of x is: 1

(a) 21 **(b)** 22 **(c)** 23 **(d)** 11

12. A lot of 180 ball pens contains 27 defective pens. One pen is drawn at random from the lot. The probability that the pen is good (not defective) is: 1

(a) $\frac{3}{20}$ **(b)** $\frac{77}{90}$ **(c)** $\frac{38}{45}$ **(d)** $\frac{17}{20}$

13. **Assertion (A):** When a solid metallic sphere is melted and recast into a cylinder, its surface area remains the same. 1

Reason (R): When a solid is melted and recast, its volume remains the same.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

14. **Assertion (A):** The modal class of a grouped frequency distribution is the class with the lowest frequency. 1

Reason (R): The mode is the value that occurs most often in the data.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

15. **Assertion (A):** The probability of an event can be 1.5. 1

Reason (R): For any event E, $0 \leq P(E) \leq 1$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. The mean of a distribution with classes 40–50, 50–60, 60–70, 70–80, 80–90, 90–100, 100–110, 110–120 and frequencies 7, 23, 19, p, 9, 16, 13, 11 respectively is 78.8. Find the value of p. 2

17. Two fair dice are thrown together and the numbers on the upper faces are noted. Find the probability that the first number is greater than the second. 2

18. A hemispherical bowl has outer radius 14 cm and inner radius 3.5 cm. Find the volume of material used to make the bowl. 2

19. The daily wages (in ₹) of workers in a factory are given below. 2

Daily wage (₹)	100–120	120–140	140–160	160–180	180–200	200–220	220–240	240–260
Number of workers	19	3	4	12	20	13	22	7

Find the mode.

20. A bag contains 16 red balls and x blue balls. The probability of drawing a blue ball is 5 times the probability of drawing a red ball. Find the value of x. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. A bag contains 13 red and 11 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{16}{27}$. Find the value of x. 3

22. A solid cylinder of base radius 12 cm and height 14 cm is melted and recast into identical solid cones, each of base radius 6 cm and height 21 cm. Find the number of cones obtained. 3

23. The median of a distribution with classes 20–30, 30–40, 40–50, 50–60, 60–70, 70–80, 80–90 and frequencies 7, 10, x, 24, y, 6, 9 respectively, and total frequency 100, is 55. Find the value of x. 3

24. A box contains 66 tickets numbered 1 to 66. One ticket is drawn at random. Find the probability that the number drawn is a composite number. 3

OR

A solid metallic cylinder of base radius 12 cm and height 24 cm is melted and recast into solid spheres of radius 6 cm. Find the number of spheres obtained. 3

25. The ages (in years) of patients admitted to a hospital in a month are given below. 3

Age (years)	100–120	120–140	140–160	160–180	180–200	200–220	220–240	240–260
Number of patients	4	12	2	21	22	21	14	4

Find the mean and the mode (in this order).

OR

Two fair dice are thrown together and the numbers on the upper faces are noted. Find the probability that at least one die shows 6. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** A bag contains 4 red, 5 blue and 6 green balls. One ball is drawn at random. 5

(i) How many equally likely outcomes are there?

(ii) Find the probability that the ball is red.

(iii) Find the probability that the ball is not green.

(iv) Find the probability that the ball is blue or green. If 5 more red balls are added to the bag, what is the new probability of drawing a red ball?

27. The daily wages (in ₹) of workers in a factory are given below. 5

Daily wage (₹)	20–30	30–40	40–50	50–60	60–70	70–80	80–90
Number of workers	18	13	5	16	23	10	15

Find the mean and the mode (in this order).

OR

A bag contains 9 red and 21 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{8}{15}$. Find the value of x. 5

END OF QUESTION PAPER Total: 50 Marks